1. Clinical Question & Core Hypothesis

Clinical Question

DDR-deficient NMIBC tumors (low ERCC-family/HR activity) should be more sensitive to cisplatin, yet some DDR-low tumors fail to achieve pathologic response. What tumor-extrinsic features modulate or blunt intrinsic DDR-related chemosensitivity? 2) Tier 1 paths used: ERCC2 (well-established cisplatin-sensitizing lesion in bladder cancer) and ERCC1 (NER mediator linked to platinum resistance). I built on these because they form the core mechanistic axis tying DNA-repair capacity to cisplatin cytotoxicity. 3) Tier 2 supporting evidence: HR genes (BRCA1/2, RAD51), broader NER/DSB sensors (ATM, CHEK1, ERCC4/5), and immune-context markers (CD274, CD8A) can modify tumor cell sensitivity or microenvironmental elimination; importantly, stromal barriers (TSR) are mechanistically plausible modulators of drug delivery and pro-survival signaling. 4) Endpoint choice: therapy_response (pathologic response / downstaging after cisplatin-based neoadjuvant chemotherapy) is the direct clinical question the IMED score is designed to predict and is the most actionable endpoint for treatment selection. 5) Deliberate exclusions: I excluded TIL/TLS as core imaging features to preserve focus on a delivery/barrier hypothesis (TSR chosen instead); I limited anchor genes to ERCC2 and ERCC1 to remain mechanistically sharp and put other DDR/immune genes into secondary_modifiers for exploratory adjustment and model refinement?

Working Hypothesis

In DDR-deficient NMIBC, low ERCC2 and low ERCC1 expression increase cisplatin sensitivity, but this sensitivity is attenuated in tumors with a high stromal-to-tumor ratio (TSR) due to reduced intratumoral drug delivery and stromal-derived pro-survival signaling; clinically, patients with low ERCC2/ERCC1 and low TSR will have higher therapy_response to cisplatin-based neoadjuvant therapy, whereas patients with low ERCC2/ERCC1 but high TSR will exhibit diminished responses, indicating a significant ERCC1/ERCC2 × TSR interaction that predicts cisplatin outcomes.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	ERCC2, ERCC1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

ERCC2: ERCC2 and ERCC1 to remain mechanistically sharp and put other DDR/immune genes into secondary_modifiers for exploratory adjustment and model refinement.

ERCC1: ERCC1 to remain mechanistically sharp and put other DDR/immune genes into secondary_modifiers for exploratory adjustment and model refinement.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	BRCA1	Supporting modifier from knowledge graph
Secondary	BRCA2	Supporting modifier from knowledge graph
Secondary	RAD51	Supporting modifier from knowledge graph
Secondary	ATM	Supporting modifier from knowledge graph
Secondary	CHEK1	Supporting modifier from knowledge graph
Secondary	ERCC4	Supporting modifier from knowledge graph
Secondary	ERCC5	Supporting modifier from knowledge graph
Secondary	PALB2	Supporting modifier from knowledge graph
Secondary	FANCD2	Supporting modifier from knowledge graph
Secondary	MRE11A	Supporting modifier from knowledge graph
Secondary	NBN	Supporting modifier from knowledge graph
Secondary	CD274	Supporting modifier from knowledge graph

Secondary	CD8A	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: and a single imaging feature (TSR) creates a focused mechanistic axis (DDR deficiency plus delivery barrier) to test a clear interaction effect on a clinically meaningful endpoint (therapy_response). Secondary modifiers (Tier 2 DDR/immune genes) are acknowledged as potential refiners of the effect size and direction but are not required for the primary hypothesis.

Prediction 2: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- BRCA1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- BRCA2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.